

Fermi-Dirac Distribution Limits and Ratio

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User prompt: analyse the expression: $\gamma(T) = f(E_i / T) \cdot f(-E_j / T)$, where E_i and E_j are some energy, $f(x)$ is the fermi dirac distribution. Analyse the ratio: $\gamma(T_1) / \gamma(T_2)$ for $T_1 \rightarrow \infty$ and $T_2 \rightarrow 0$.

Response: To analyze the ratio of $\gamma(T_1) / \gamma(T_2)$ at the specified limits, we evaluate the standard Fermi-Dirac distribution function, $f(x) = \frac{1}{e^{x+1} + 1}$, where the argument is scaled energy $x = E/T$ (assuming the Boltzmann constant $k_B \equiv 1$ and E is measured relative to the chemical potential μ).

The expression given is:

$$\gamma(T) = f\left(\frac{E_i}{T}\right) \cdot f\left(-\frac{E_j}{T}\right) = \left(\frac{1}{e^{E_i/T} + 1}\right) \left(\frac{1}{e^{-E_j/T} + 1}\right)$$

Note that $f(-x) = 1 - f(x)$, which physically represents the probability of an unoccupied state (a "hole") at energy E_j .

1. The High-Temperature Limit ($T_1 \rightarrow \infty$)

As the temperature approaches infinity, the thermal energy overwhelms any specific energy level. The arguments of the Fermi-Dirac functions approach zero regardless of the values of E_i and E_j :

$$\lim_{T_1 \rightarrow \infty} \frac{E_i}{T_1} = 0 \quad \text{and} \quad \lim_{T_1 \rightarrow \infty} -\frac{E_j}{T_1} = 0$$

Evaluating the distribution function at $x = 0$:

$$f(0) = \frac{1}{e^0 + 1} = \frac{1}{2}$$

Thus, the asymptotic value for γ is:

$$\gamma(T_1 \rightarrow \infty) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$$

2. The Low-Temperature Limit ($T_2 \rightarrow 0$)

As the temperature approaches absolute zero, the Fermi-Dirac distribution converges to the Heaviside step function. The limit depends strictly on whether the energies are above or below the Fermi level (chemical potential, which is 0 in this relative scale):

$$\lim_{T_2 \rightarrow 0} f\left(\frac{E}{T_2}\right) = \begin{cases} 0 & \text{if } E > 0 \\ 1 & \text{if } E < 0 \end{cases}$$

Applying this to both terms in $\gamma(T_2)$:

- $f(E_i/T_2) \rightarrow 1$ if $E_i < 0$, and 0 if $E_i > 0$.
- $f(-E_j/T_2) \rightarrow 1$ if $-E_j < 0$ (meaning $E_j > 0$), and 0 if $-E_j > 0$ (meaning $E_j < 0$).

Therefore, the limit of $\gamma(T_2)$ evaluates to a binary outcome based on the signs of E_i and E_j :

$$\gamma(T_2 \rightarrow 0) = \begin{cases} 1 & \text{if } E_i < 0 \text{ and } E_j > 0 \\ 0 & \text{otherwise (if } E_i > 0 \text{ or } E_j < 0) \end{cases}$$

3. The Ratio $\gamma(T_1)/\gamma(T_2)$

Combining the high- and low-temperature limits yields the ratio:

$$R = \frac{\gamma(T_1 \rightarrow \infty)}{\gamma(T_2 \rightarrow 0)} = \frac{1/4}{\gamma(T_2 \rightarrow 0)}$$

The behavior of this ratio depends entirely on the initial state energies:

Case A: $E_i < 0$ and $E_j > 0$ The low-temperature state is fully occupied (the electron state E_i is filled and the hole state E_j is available).

$$R = \frac{1/4}{1} = \frac{1}{4}$$

Case B: $E_i > 0$ or $E_j < 0$ The low-temperature limit evaluates to zero, meaning the joint probability is completely suppressed by the Fermi sea step. The ratio diverges to infinity. To quantify the divergence as T_2 becomes infinitesimally small:

- If $E_i > 0, E_j > 0$: $\gamma(T_2) \approx e^{-E_i/T_2} \implies R \propto e^{E_i/T_2} \rightarrow \infty$
- If $E_i < 0, E_j < 0$: $\gamma(T_2) \approx e^{E_j/T_2} \implies R \propto e^{-E_j/T_2} \rightarrow \infty$
- If $E_i > 0, E_j < 0$: $\gamma(T_2) \approx e^{-(E_i-E_j)/T_2} \implies R \propto e^{(E_i-E_j)/T_2} \rightarrow \infty$

User prompt: How about the ratio: $(\gamma_{ij}(T_1) + \gamma_{ij}(T_2)) / (\gamma_{ji}(T_1) + \gamma_{ji}(T_2))$

Response: To analyze this new ratio, we first define the reverse process term, $\gamma_{ji}(T)$,

which swaps the energy states E_i and E_j :

$$\gamma_{ji}(T) = f\left(\frac{E_j}{T}\right) \cdot f\left(-\frac{E_i}{T}\right)$$

Using the limits established previously for $T_1 \rightarrow \infty$ and $T_2 \rightarrow 0$, we evaluate the components of the ratio:

$$R' = \frac{\gamma_{ij}(T_1 \rightarrow \infty) + \gamma_{ij}(T_2 \rightarrow 0)}{\gamma_{ji}(T_1 \rightarrow \infty) + \gamma_{ji}(T_2 \rightarrow 0)}$$

1. High-Temperature Terms ($T_1 \rightarrow \infty$)

At infinite temperature, the Fermi-Dirac distribution is strictly $1/2$ for all finite energy states. Both the forward and reverse terms converge to the same value:

$$\gamma_{ij}(T_1) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$$

$$\gamma_{ji}(T_1) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$$

2. Low-Temperature Terms ($T_2 \rightarrow 0$)

At absolute zero, the terms evaluate to 1 only if the first energy state is below the Fermi level (filled) and the second is above (empty). Otherwise, they evaluate to 0:

- $\gamma_{ij}(T_2) = 1$ if $E_i < 0$ and $E_j > 0$, else 0.
- $\gamma_{ji}(T_2) = 1$ if $E_j < 0$ and $E_i > 0$, else 0.

3. Evaluating the Ratio

Substituting the high-temperature constants into the ratio gives:

$$R' = \frac{1/4 + \gamma_{ij}(T_2)}{1/4 + \gamma_{ji}(T_2)}$$

Because the low-temperature terms can only be 0 or 1, we can map out the ratio across all possible combinations of energy state signs:

State Energies	$\gamma_{ij}(T_2)$	$\gamma_{ji}(T_2)$	Ratio R'	Result
$E_i < 0$ and $E_j > 0$	1	0	$\frac{1/4+1}{1/4+0}$	5

State Energies	$\gamma_{ij}(T_2)$	$\gamma_{ji}(T_2)$	Ratio R'	Result
$E_i > 0$ and $E_j < 0$	0	1	$\frac{1/4+0}{1/4+1}$	1/5
Both $E > 0$	0	0	$\frac{1/4+0}{1/4+0}$	1
Both $E < 0$	0	0	$\frac{1/4+0}{1/4+0}$	1

By adding the infinite-temperature baseline, the zero-temperature divergence is eliminated. The ratio is bounded and perfectly symmetric, yielding 5, 1/5, or 1 depending strictly on which sides of the Fermi level the two energy states lie.

User prompt: what about: let $\gamma_{ij}(T) = (1+N(\omega, T))\delta(E_j - E_i + \omega) + N(\omega, T)\delta(E_j - E_i - \omega)$, and the ratio $(\gamma_{ij}(T_1) + \gamma_{ij}(T_2)) / (\gamma_{ji}(T_1) + \gamma_{ji}(T_2))$, where δ is dirac delta and $N(\omega, T)$ is the Bose distribution, with $(\omega = \text{omega here, Its just faster to write for me})$

Response: To analyze this ratio with **Bose-Einstein statistics**, we first determine the reverse transition rate, $\gamma_{ji}(T)$, by swapping indices $i \leftrightarrow j$.

Using the property of the Dirac delta function $\delta(-x) = \delta(x)$:

$$\gamma_{ji}(T) = [1 + N(\omega, T)] \delta(E_j - E_i - \omega) + N(\omega, T) \delta(E_j - E_i + \omega)$$

Here, $N(\omega, T) = \frac{1}{e^{\omega/T} - 1}$ is the average Bose occupation number (assuming $\omega > 0$).

1. Temperature Limits of Bose Occupation

- **Low Temperature ($T_2 \rightarrow 0$):** The thermal photon/boson field freezes out completely:

$$\lim_{T_2 \rightarrow 0} N(\omega, T_2) = 0 \implies 1 + N(\omega, T_2) \rightarrow 1$$

Thus, $\gamma_{ij}(T_2) = \delta(E_j - E_i + \omega)$ (only spontaneous emission survives).

- **High Temperature ($T_1 \rightarrow \infty$):** By Taylor expanding $e^{\omega/T_1} \approx 1 + \frac{\omega}{T_1}$, the Bose occupation number diverges linearly with temperature:

$$N(\omega, T_1) \approx \frac{T_1}{\omega} \rightarrow \infty$$

Because $N(\omega, T_1) \gg 1$, stimulated processes completely dominate:

$$1 + N(\omega, T_1) \approx N(\omega, T_1) \approx \frac{T_1}{\omega}$$

Therefore, at $T_1 \rightarrow \infty$, the forward and reverse rates become equal:

$$\gamma_{ij}(T_1) \approx \gamma_{ji}(T_1) \approx \frac{T_1}{\omega} [\delta(E_j - E_i + \omega) + \delta(E_j - E_i - \omega)]$$

2. Evaluating the Ratio

To get a non-zero result, the energy difference between state i and state j must match the quantum energy ω .

Case 1: Downward Transition ($E_i - E_j = \omega$)

Here, $\delta(E_j - E_i + \omega) = \delta(0)$ is active, while the absorption delta is 0.

$$\gamma_{ij}(T) = [1 + N(\omega, T)]\delta(0)$$

$$\gamma_{ji}(T) = N(\omega, T)\delta(0)$$

Substituting into the ratio for any T_1 and T_2 :

$$R = \frac{\gamma_{ij}(T_1) + \gamma_{ij}(T_2)}{\gamma_{ji}(T_1) + \gamma_{ji}(T_2)} = \frac{[1 + N(T_1)] + [1 + N(T_2)]}{N(T_1) + N(T_2)}$$

Taking the limits $T_2 \rightarrow 0 \implies N(T_2) \rightarrow 0$:

$$R = \frac{N(T_1) + 2}{N(T_1)} = 1 + \frac{2}{N(T_1)}$$

Taking $T_1 \rightarrow \infty \implies N(T_1) \rightarrow \infty$:

$$\lim_{T_1 \rightarrow \infty} R = 1$$

Case 2: Upward Transition ($E_j - E_i = \omega$)

Here, $\delta(E_j - E_i - \omega) = \delta(0)$ is active.

$$\gamma_{ij}(T) = N(\omega, T)\delta(0)$$

$$\gamma_{ji}(T) = [1 + N(\omega, T)]\delta(0)$$

Substituting into the ratio:

$$R = \frac{N(T_1) + N(T_2)}{[1 + N(T_1)] + [1 + N(T_2)]}$$

Taking $T_2 \rightarrow 0 \implies N(T_2) \rightarrow 0$:

$$R = \frac{N(T_1)}{N(T_1) + 2}$$

Taking $T_1 \rightarrow \infty \implies N(T_1) \rightarrow \infty$:

$$\lim_{T_1 \rightarrow \infty} R = 1$$

Summary & Key Comparison to Fermi-Dirac

Energy Condition	Bose-Einstein Ratio ($T_1 \rightarrow \infty, T_2 \rightarrow 0$)
$ E_i - E_j = \omega$	1
$ E_i - E_j \neq \omega$	Undefined (0/0)

Why Bose-Einstein gives 1 vs. Fermi-Dirac's 5 (or 1/5):

- Pauli Exclusion (Fermi-Dirac):** The occupation probability is strictly bounded ($0 \leq f \leq 1$), approaching $1/2$ at $T \rightarrow \infty$. Because the high-temperature term remains finite, the low-temperature term $f(T_2 \rightarrow 0)$ retains significant weight in the sum, preserving a directional preference (yielding ratios of 5 or $1/5$).
- No Exclusion Principle (Bose-Einstein):** Unbounded photon/boson accumulation causes $N(\omega, T_1)$ to blow up as $T_1 \rightarrow \infty$. High-temperature stimulated emission and absorption completely overwhelm the spontaneous emission term at T_2 , making forward and reverse rates symmetric and driving the ratio to **1**.

User prompt: What if you replaced $N(w, T)$ with $f(w, T)$ and calculated the ratio?

Response: Replacing the Bose-Einstein distribution $N(\omega, T)$ with the Fermi-Dirac distribution $f(\omega, T) = \frac{1}{e^{\omega/T} + 1}$ gives a ratio of 5 (for a downward transition) or $1/5$ (for an upward transition).

This occurs because Fermi-Dirac statistics saturate to a finite value at infinite temperature rather than diverging.

1. Temperature Limits of $f(\omega, T)$

Assuming $\omega > 0$:

- **Low Temperature ($T_2 \rightarrow 0$):**

$$\lim_{T_2 \rightarrow 0} f(\omega, T_2) = 0 \implies 1 + f(\omega, T_2) \rightarrow 1$$

$$\circ \gamma_{ij}(T_2) = \delta(E_j - E_i + \omega)$$

$$\circ \gamma_{ji}(T_2) = \delta(E_j - E_i - \omega)$$

- **High Temperature** ($T_1 \rightarrow \infty$): Since $e^{\omega/T_1} \rightarrow 1$, the distribution approaches $1/2$:

$$\lim_{T_1 \rightarrow \infty} f(\omega, T_1) = \frac{1}{1+1} = \frac{1}{2} \implies 1 + f(\omega, T_1) \rightarrow \frac{3}{2}$$

$$\circ \gamma_{ij}(T_1) = \frac{3}{2}\delta(E_j - E_i + \omega) + \frac{1}{2}\delta(E_j - E_i - \omega)$$

$$\circ \gamma_{ji}(T_1) = \frac{3}{2}\delta(E_j - E_i - \omega) + \frac{1}{2}\delta(E_j - E_i + \omega)$$

2. Calculating the Ratio

$$R = \frac{\gamma_{ij}(T_1) + \gamma_{ij}(T_2)}{\gamma_{ji}(T_1) + \gamma_{ji}(T_2)}$$

Case 1: Downward Transition ($E_i - E_j = \omega$)

The emission delta function $\delta(E_j - E_i + \omega) = \delta(0)$ is active:

$$R = \frac{\gamma_{ij}(T_1) + \gamma_{ij}(T_2)}{\gamma_{ji}(T_1) + \gamma_{ji}(T_2)} = \frac{3/2 + 1}{1/2 + 0} = \frac{5/2}{1/2} = 5$$

Case 2: Upward Transition ($E_j - E_i = \omega$)

The absorption delta function $\delta(E_j - E_i - \omega) = \delta(0)$ is active:

$$R = \frac{\gamma_{ij}(T_1) + \gamma_{ij}(T_2)}{\gamma_{ji}(T_1) + \gamma_{ji}(T_2)} = \frac{1/2 + 0}{3/2 + 1} = \frac{1/2}{5/2} = \frac{1}{5}$$

Summary Comparison

Distribution Used	$T_1 \rightarrow \infty$ Saturation	Downward Transition Ratio R
Bose-Einstein $N(\omega, T)$	$N \rightarrow \infty$ (unbounded accumulation)	1
Direct Substitution $f(\omega, T)$	$f \rightarrow 1/2$	5

Distribution Used	$T_1 \rightarrow \infty$ Saturation	Downward Transition Ratio R
Physical Fermi-Blocking ($1 - f(\omega, T)$)	$(1 - f) \rightarrow 1/2$	3

Note on Fermi-Blocking: If the factor $(1 + N)$ is modified to represent Pauli blocking for fermions, $(1 - f(\omega, T))$, the high-temperature emission factor becomes $1 - 1/2 = 1/2$. Substituting this into the ratio yields $\frac{1/2+1}{1/2+0} = \mathbf{3}$.

User prompt: So in the fermi cases the cold bath is still relevant as T_1 grows (not necessarily infinite, but larger) and in the Bose case the cold bath does not contribute at very low T_2 ?

Response: Your intuition is spot on. The fundamental difference lies in **saturation vs. divergence** at high temperatures.

1. The Fermi Case: Why the Cold Bath Stays Relevant

For fermions, the Pauli exclusion principle enforces a strict ceiling on occupation probabilities ($f \leq 1$).

Even if $T_1 \rightarrow \infty$, the distribution function saturates at $f(\omega, T_1) \rightarrow 1/2$. Because the hot bath's contribution cannot grow beyond this limit:

- **Hot bath emission contribution:** $\approx 1/2$ (or $3/2$ without Pauli blocking)
- **Cold bath spontaneous emission ($T_2 \rightarrow 0$):** 1

Because 1 and $1/2$ are the same order of magnitude, **the cold bath's spontaneous emission accounts for a massive fraction (40%–66%) of the total downward transitions**, no matter how hot T_1 becomes. The hot bath cannot "drown out" the cold bath.

2. The Bose Case: Why the Hot Bath Drowns Out the Cold Bath

For bosons, there is no exclusion principle. Photons/bosons can pile into the same mode indefinitely.

As T_1 increases, $N(\omega, T_1) \approx \frac{T_1}{\omega}$ grows without bound:

$$\text{Ratio } R = \frac{\gamma_{ij}(T_1) + \gamma_{ij}(T_2)}{\gamma_{ji}(T_1) + \gamma_{ji}(T_2)} = \frac{[N(T_1) + 1] + 1}{N(T_1) + 0} = 1 + \frac{2}{N(T_1)}$$

- At $T_2 \rightarrow 0$, the cold bath *does* contribute its 1 unit of spontaneous emission.

- However, if T_1 is large, $N(T_1)$ becomes huge (e.g., $N(T_1) = 10,000$).
- The cold bath's spontaneous emission (+1) is completely negligible compared to the massive flood of stimulated emissions and absorptions driven by the hot bath ($N(T_1)$).

Summary Analogy

- **Fermions:** Turning on a hot bath is like turning on a flashlight next to a candle (the cold bath). Because the flashlight has a maximum brightness cap (Pauli ceiling), the candle's flame remains clearly visible and significant to the overall light.
 - **Bosons:** Turning on a hot bath is like turning on a stadium floodlight next to a candle. As T_1 grows, the floodlight gets infinitely brighter, completely blinding out the candle's contribution.
-